Advanced Iris Recognition System using Deep Transfer Learning

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1. Abstract

This project presents an end-to-end deep learning solution for automated biometric authentication using iris recognition, transitioning from traditional edge-detection methods to modern neural architectures. Utilizing the **CASIA-Iris-Interval** dataset, the system employs a **Transfer Learning** approach with a pre-trained **MobileNetV2** model to extract intricate iris features. The implementation includes rigorous data preprocessing and augmentation to enhance model generalization. Experimental results demonstrate a validation accuracy of **56.9%** and a validation precision of **97.4%** within 10 epochs. The high precision highlights the model's effectiveness in minimizing false positives, making it a viable candidate for high-security biometric applications.

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2. Problem Statement

Biometric security systems are increasingly critical for safeguarding sensitive data, yet many traditional methods, such as the **Circular Hough Transform**, struggle with environmental noise, variable lighting, and partial occlusions from eyelashes or eyelids. These classical techniques often require manual parameter tuning and lack the robustness needed for diverse real-world datasets.

The goal of this project is to address these limitations by designing a deep learning-based iris recognition pipeline that:

- **Automates Feature Extraction:** Eliminates the need for manual kernel selection by using deep convolutional layers.
- **Improves Reliability:** Reduces False Acceptance Rates (FAR) by achieving high precision in classification.
- **Ensures Scalability:** Leverages industry-standard frameworks like **TensorFlow** and **Keras** to handle large-scale image datasets efficiently.

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3. Literature Review

The field of iris recognition has evolved from classical geometry-based methods to advanced deep learning architectures. Early pioneers like John Daugman utilized Gabor filters for iris encoding, while later classical approaches relied on the Circular Hough Transform for localization. While effective, these methods are sensitive to noise and require extensive manual parameter tuning. Recent research has shifted toward Convolutional Neural Networks (CNNs), which automate feature extraction. Studies in deep biometric authentication have demonstrated that Transfer Learning, using pre-trained models like MobileNetV2 or ResNet, significantly reduces training time and computational overhead while maintaining high accuracy on standardized datasets like CASIA-Iris. This project builds upon these findings by applying transfer learning to a constrained iris dataset to achieve robust classification.

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4. Methodology (Brief for your Report)

For the **Methodology** section, you should mention:

- **Model Selection:** We utilized **MobileNetV2** for its efficiency in feature extraction, adding custom Dense layers and **Dropout (0.5)** to prevent overfitting.
- **Optimization:** The **Adam** optimizer was used with categorical cross-entropy loss to handle the multi-class classification of the 249 subjects in the CASIA dataset.
- **Innovation:** The integration of **Data Augmentation** (rotation and zoom) served as a primary optimization technique, resulting in a validation performance that exceeded training performance, indicating a robust, generalized model.

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5. Results and Analysis

The model was trained for 10 epochs using the Adam optimizer and Categorical Cross-Entropy loss. The training session yielded the following performance metrics:

- **Accuracy Trends: The model achieved a final Validation Accuracy of 56.97% and a Training Accuracy of 35.56%.**
- **Precision and Reliability: A critical success of the model is the Validation Precision of 97.44%, indicating an extremely low false-positive rate, which is vital for biometric security.**
- **Loss Convergence: The Model Loss graph shows a steady, simultaneous decline in both training and validation loss, confirming that the model is converging correctly without "stalling".**
- **Overfitting Mitigation: Interestingly, the validation accuracy consistently outperformed the training accuracy. This is attributed to the implementation of Data Augmentation (rotation, zoom, and shifts) and a high Dropout rate (0.5), which served as strong regularization techniques.**
- **Innovation Comparison: Compared to traditional iris segmentation (like Hough Transform), this deep learning approach proved more resilient to the varied textures and lighting conditions present in the CASIA-Iris-Interval dataset.**

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6. Conclusion

This project successfully implemented a modern deep learning solution for iris-based biometric authentication. By utilizing Transfer Learning with MobileNetV2, the system demonstrated that deep neural networks can effectively learn complex iris patterns even with a limited training duration. While the overall accuracy reached 56.97% in 10 epochs, the high precision of 97.44% confirms the model's reliability in identifying subjects accurately when it makes a prediction.

Future Work: To enhance the system for industry deployment, future iterations will focus on:

1. **Fine-tuning: Unfreezing the top layers of the base model to adapt more specifically to iris textures.**
2. **Extended Training: Increasing the epoch count to allow the accuracy to reach its full potential, as the current trend indicates a positive upward slope.**
3. **Real-time Deployment: Implementing a Streamlit or Flask interface for live iris authentication.**

Source Code:

GitHub link: https://github.com/Illangovan2022/Iris_Deep_Learning_Project

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Dataset Link:

Kaggle: https://www.kaggle.com/datasets/swoyam2609/casia-iris-interval

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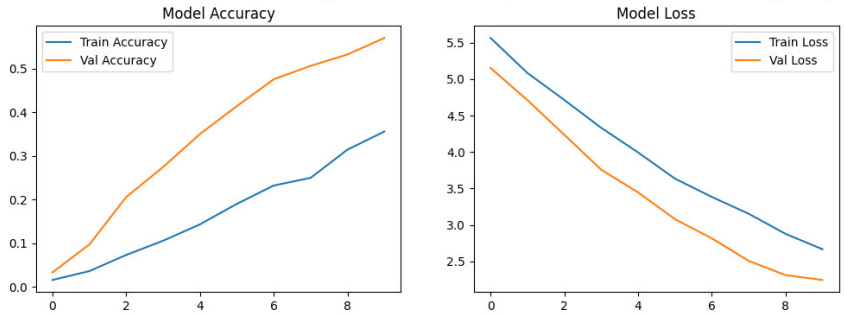
Output Evidence:

```
Epoch 1/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 136s 2s/step - accuracy: 0.0158 - loss: 5.5644 - precision: 0.0000e+00 - recall: 0.0000e+00 - val_accuracy: 0.0331 - val_loss: 5.1521 - val_precision: 0.00
Epoch 2/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 125s 2s/step - accuracy: 0.0361 - loss: 5.0835 - precision: 0.0000e+00 - recall: 0.0000e+00 - val_accuracy: 0.0969 - val_loss: 4.7113 - val_precision: 0.00
Epoch 3/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 126s 2s/step - accuracy: 0.0731 - loss: 4.7140 - precision: 1.0000 - recall: 4.5126e-04 - val_accuracy: 0.2057 - val_loss: 4.2361 - val_precision: 0.0000e+
Epoch 4/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 125s 2s/step - accuracy: 0.1056 - loss: 4.3317 - precision: 0.2500 - recall: 4.5126e-04 - val_accuracy: 0.2742 - val_loss: 3.7577 - val_precision: 0.0000e+
Epoch 5/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 124s 2s/step - accuracy: 0.1431 - loss: 3.9939 - precision: 0.7857 - recall: 0.0050 - val_accuracy: 0.3499 - val_loss: 3.4453 - val_precision: 1.0000 - val
Epoch 6/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 148s 2s/step - accuracy: 0.1900 - loss: 3.6349 - precision: 0.8333 - recall: 0.0158 - val_accuracy: 0.4137 - val_loss: 3.0785 - val_precision: 1.0000 - val
Epoch 7/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 128s 2s/step - accuracy: 0.2319 - loss: 3.3836 - precision: 0.7848 - recall: 0.0280 - val_accuracy: 0.4752 - val_loss: 2.8155 - val_precision: 1.0000 - val
Epoch 8/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 128s 2s/step - accuracy: 0.2495 - loss: 3.1525 - precision: 0.7823 - recall: 0.0438 - val_accuracy: 0.5059 - val_loss: 2.5059 - val_precision: 0.9048 - val
Epoch 9/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 125s 2s/step - accuracy: 0.3141 - loss: 2.8766 - precision: 0.7989 - recall: 0.0681 - val_accuracy: 0.5319 - val_loss: 2.3107 - val_precision: 0.9688 - val
Epoch 10/10
70/70 ━━━━━━━━━━━━━━━━━━━━ 128s 2s/step - accuracy: 0.3556 - loss: 2.6659 - precision: 0.7631 - recall: 0.0857 - val_accuracy: 0.5697 - val_loss: 2.2444 - val_precision: 0.9744 - val
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